An Onsager Variational Scheme for Pressure-Driven Tumor Growth and Hele–Shaw Limits

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Abstract

Pressure-driven tumor growth models describe the coupling between cell proliferation and mechanical pressure and naturally lead to moving free boundary problems. Their numerical approximation is challenging due to degenerate diffusion, pressure-dependent proliferation, and the stiffness of the pressure law $p = n^\gamma$ for large γ. In this paper, we propose a structure-preserving finite difference method for this class of pressure-driven tumor growth models with pressure-dependent proliferation. The method is derived from the Onsager variational principle. The key idea is to introduce a modified energy shifted by the homeostatic pressure, so that the growth term can be written in a dissipative form and incorporated together with the transport part into a unified Rayleighian formulation. This formulation leads to a time-discrete constrained minimization problem and a fully discrete scheme with explicit mobilities and an implicit pressure update. We prove that the scheme preserves nonnegativity and the homeostatic upper bound, satisfies a discrete modified energy dissipation law, and admits a fixed-grid stiff-pressure limiting structure. Numerical experiments in one and two spatial dimensions demonstrate the accuracy of the method, its convergence toward the Hele–Shaw limit for large γ, and its ability to capture free boundary evolutions with topology changes.

Key words. Pressure-driven tumor growth, Onsager variational principle, structure-preserving scheme, energy stability, Hele–Shaw limit.

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1 Introduction

Living tissues are active materials in which proliferation, mechanical pressure, and crowding effects are strongly coupled [1, 11, 25]. Continuum models for such mechanobiological processes must describe both bulk evolution and the motion of free boundaries separating occupied and empty regions, and, in the stiff-pressure regime, they naturally lead to singular limits in which strong mechanical resistance enforces an incompressibility constraint in densely packed tissues [22, 23, 24]. Tumor growth is a prototypical setting in which these mechanisms appear. For this reason, pressure-driven tumor growth models have become a useful class of problems for studying the connection between degenerate diffusion, free boundary motion, and reliable numerical simulation.

In this paper, we consider the pressure-driven tumor growth model [23]

$$\partial_t n + \nabla \cdot j = nG(p), \qquad j = -n\nabla p, \qquad p = n^\gamma, \tag{1.1}$$

posed in a bounded domain $\Omega \subset \mathbb{R}^d$. Here $n = n(\mathbf{x}, t) \geq 0$ denotes the tumor cell density, $j = j(\mathbf{x}, t)$ is the cell flux, $p = p(\mathbf{x}, t)$ is the mechanical pressure, and $\gamma > 1$ is the stiffness

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parameter in the pressure law. The function G is the net proliferation rate. We assume pressure-inhibited proliferation with a homeostatic pressure $p_H > 0$ [1, 22, 23], namely

$$G(p_H) = 0, \qquad G'(p) \le 0, \quad p \ge 0. \tag{1.2}$$

This implies

$$(p - p_H)G(p) \le 0, \qquad p \ge 0, \tag{1.3}$$

which is the key structural condition behind the dissipative formulation used in this work.

For fixed γ, (1.1) is a degenerate reaction-diffusion equation of porous-medium type. The degeneracy gives finite speed of propagation and allows the formation of sharp moving interfaces. As $\gamma \to \infty$, the pressure law enforces a congestion constraint and the pressure becomes a Lagrange multiplier for the saturated region. Formally, provided the corresponding bounds hold initially, the limiting density and pressure satisfy [6, 23]

$$0 \le n_\infty \le 1, \qquad 0 \le p_\infty \le p_H, \qquad p_\infty(1 - n_\infty) = 0, \tag{1.4}$$

together with the complementarity relation

$$p_\infty(\Delta p_\infty + G(p_\infty)) = 0. \tag{1.5}$$

In the patch case $n_\infty = \mathbf{1}_{\Omega(t)}$, with $\Omega(t) = \{p_\infty > 0\}$, this gives the Hele–Shaw free boundary problem

$$\begin{cases} -\Delta p_\infty = G(p_\infty), & \text{in } \Omega(t), & (1.6\text{a}) \\ p_\infty = 0, & \text{on } \partial\Omega(t), & (1.6\text{b}) \\ V_n = -\partial_\nu p_\infty, & \text{on } \partial\Omega(t). & (1.6\text{c}) \end{cases}$$

Here V_n is the outward normal velocity of the free boundary and ν is the outward unit normal to $\Omega(t)$.

The mathematical analysis of these models has developed rapidly. B. Perthame, F. Quirós, and J. L. Vázquez established the Hele–Shaw asymptotics for mechanical tumor growth models and derived the complementarity relation in the incompressible limit [23]. I. Kim and N. Požár proved convergence from the porous medium equation to Hele–Shaw flow for general initial densities [14]. B. Perthame and N. Vauchelet studied the incompressible limit of a mechanical tumor growth model with viscosity [24]. B. Perthame, F. Quirós, M. Tang, and N. Vauchelet derived Hele–Shaw-type systems from cell models that include active motion [22]. N. David and B. Perthame analyzed tumor growth models coupled to nutrients, including the free boundary limit and pressure regularity [5].

From the numerical point of view, (1.1) contains several interacting difficulties. The diffusion degenerates in vacuum regions, so standard parabolic smoothing is not available near the interface. The reaction term changes the mass and must remain compatible with pressure-inhibited proliferation. For large γ, the pressure law is stiff and the numerical density should approach the congestion constraint without spurious overshoots. A useful scheme should therefore preserve nonnegativity, the homeostatic upper bound, and a suitable energy dissipation law, while also retaining a fixed-grid stiff-pressure limiting structure in the Hele–Shaw limit.

There is a large literature on numerical methods for porous medium and degenerate diffusion equations. J. L. Graveleau and P. Jamet developed early finite difference methods for nonlinear diffusion with finite speed of propagation [12]. E. DiBenedetto and D. Hoff proposed an interface tracking algorithm for the porous medium equation [7]. L. Monsaingeon designed an explicit finite difference method for one-dimensional generalized porous medium equations, with particular attention to interface tracking and hole filling [19]. Y. Liu, C.-W. Shu, and M. Zhang constructed high-order finite difference WENO schemes that reduce oscillations near sharp interfaces [18]. M. Bessemoulin-Chatard and F. Filbet developed finite volume schemes for nonlinear degenerate parabolic equations [2]. J. A. Carrillo, H. Ranetbauer, and M.-T. Wolfram used evolving diffeomorphisms to approximate nonlinear continuity equations [3]. C. Liu and Y. Wang constructed Lagrangian schemes from a discrete energetic variational approach [15].

Numerical methods have also been developed specifically for tumor growth models and their Hele–Shaw limits. J.-G. Liu, M. Tang, L. Wang, and Z. Zhou proposed an accurate front capturing scheme for tumor growth models with a free boundary limit [16]. The same authors studied tumor growth models with nutrients, connecting cell-density models to free-boundary dynamics [17]. N. David and X. Ruan studied an upwind finite difference scheme, proved stability estimates and an asymptotic-preserving property, and investigated focusing solutions numerically [6]. Most existing methods start from a direct discretization of the PDE. In contrast, the present work starts from a modified energy and an Onsager Rayleighian, and treats the transport flux and the pressure-dependent growth term in the same variational framework.

The Onsager variational principle provides a systematic way to derive dissipative dynamics from an energy and a dissipation mechanism. L. Onsager introduced the reciprocal relations and the variational structure of irreversible processes [20, 21]. M. Doi developed and promoted the Onsager principle as a modeling and approximation tool for soft matter systems [8, 9]. R. Jordan, D. Kinderlehrer, and F. Otto showed that the Fokker–Planck equation can be obtained as a minimizing movement scheme in the Wasserstein metric [13]. C. Duan, C. Liu, C. Wang, and X. Yue used an energetic variational approach for the porous medium equation [10]. H. Chen, H. Liu, and X. Xu recently presented a general Onsager-based framework for structure-preserving numerical schemes, emphasizing energy stability and constraints through discrete Rayleighian minimization [4].

The key observation of this paper is that the correct dissipative variable for pressure-inhibited proliferation is obtained by shifting the pressure energy by the homeostatic pressure. With this shift, the variational derivative is measured relative to p_H, and the pressure-inhibition condition makes the reaction contribution compatible with energy dissipation. This allows the transport flux and the reaction source to be incorporated into a single Rayleighian formulation. Based on this structure, we derive an Onsager variational formulation for pressure-driven tumor growth with pressure-inhibited proliferation, construct a time-discrete scheme from a discrete Rayleighian, and prove its modified energy stability. We then propose a fully discrete finite difference scheme with explicit mobilities and an implicit pressure update, and show that it preserves nonnegativity, the homeostatic upper bound, and a discrete modified energy dissipation law. We also establish a fixed-grid stiff-pressure limiting structure as $\gamma \to \infty$, including a discrete density-pressure complementarity relation and a Hele–Shaw-type pressure equation in saturated cells. Finally, one- and two-dimensional numerical experiments are presented to assess the accuracy, structure preservation, large-γ behavior, and ability of the method to capture free boundary evolutions with topology changes.

The rest of the paper is organized as follows. In Section 2, we present the model, state the assumptions on the growth function, and collect several structural properties. In Section 3, we develop the Onsager-based numerical scheme and analyze its structure-preserving properties and fixed-grid stiff-pressure limiting structure. Numerical experiments are reported in Section 4. Finally, conclusions are given in Section 5.

2 Basic properties of the model (1.1)

In this section, we recall the structural properties of the continuous model. Throughout this section, $n = n(\mathbf{x}, t)$, $j = j(\mathbf{x}, t)$, and $p = p(\mathbf{x}, t)$ are as in (1.1), and the equations are considered in $\Omega \times (0, T]$. We use the initial condition $n(\mathbf{x}, 0) = n_0(\mathbf{x})$, $\mathbf{x} \in \Omega$, and the no-flux boundary condition

$$j \cdot \nu = 0, \qquad \mathbf{x} \in \partial\Omega, \quad 0 < t \leq T. \tag{2.1}$$

Here ν is the outward unit normal vector of $\partial\Omega$. For the growth function G in (1.1), we assume that it satisfies (1.2)-(1.3). With this assumption, the following elementary estimates hold for smooth solutions.

Proposition 2.1 (Positivity, upper bound, and mass estimate)**.** *Assume that*

$$0 \le p_0(\mathbf{x}) = n_0(\mathbf{x})^\gamma \le p_H \qquad \text{in } \Omega.$$

Then any smooth solution of (1.1) *subject to* (2.1) *satisfies*

$$0 \le p(\mathbf{x},t) \le p_H, \qquad 0 \le n(\mathbf{x},t) \le n_H := p_H^{1/\gamma}, \qquad \mathbf{x} \in \Omega, \quad 0 \le t \le T.$$

Moreover,

$$\int_\Omega n(\mathbf{x},t)\,d\mathbf{x} \le e^{G(0)t} \int_\Omega n_0(\mathbf{x})\,d\mathbf{x}, \qquad 0 \le t \le T. \tag{2.2}$$

Proof. Differentiating $p = n^\gamma$ and using (1.1), we obtain

$$\partial_t p = |\nabla p|^2 + \gamma p\big(\Delta p + G(p)\big). \tag{2.3}$$

The constants 0 and p_H are respectively a subsolution and a supersolution of this equation, since $G(p_H) = 0$. Hence the comparison principle gives $0 \le p \le p_H$. The bound $0 \le n \le n_H$ follows from $p = n^\gamma$.

For the mass estimate, integrating the density equation and using the no-flux boundary condition gives

$$\frac{d}{dt}\int_\Omega n\,d\mathbf{x} = \int_\Omega nG(p)\,d\mathbf{x} \le G(0)\int_\Omega n\,d\mathbf{x},$$

where we used $p \ge 0$, $n \ge 0$, and the monotonicity of G. Gronwall's inequality then yields (2.2). ☐

Proposition 2.2 (Modified energy dissipation)**.** *Define the modified energy by*

$$\mathcal{E}(n) = \int_\Omega \left(\frac{1}{\gamma+1} n^{\gamma+1} - p_H n\right) d\mathbf{x}. \tag{2.4}$$

Let n be a nonnegative smooth solution of (1.1) *subject to the no-flux boundary condition* (2.1)*. Then*

$$\frac{d}{dt}\mathcal{E}(n) = -\int_\Omega n|\nabla p|^2\,d\mathbf{x} + \int_\Omega n(p-p_H)G(p)\,d\mathbf{x} \le 0. \tag{2.5}$$

Proof. Since

$$\frac{\delta\mathcal{E}}{\delta n} = n^\gamma - p_H = p - p_H,$$

we multiply the density equation by $p - p_H$ and use the no-flux boundary condition to obtain

$$\begin{aligned}\frac{d}{dt}\mathcal{E}(n) &= \int_\Omega (p-p_H)\partial_t n\,d\mathbf{x}\\ &= -\int_\Omega n|\nabla p|^2\,d\mathbf{x} + \int_\Omega n(p-p_H)G(p)\,d\mathbf{x}.\end{aligned}$$

The boundary term vanishes because $n\nabla p\cdot\nu = -j\cdot\nu = 0$ on $\partial\Omega$. Since $n \ge 0$ and $(p-p_H)G(p) \le 0$ by (1.3), the right-hand side is nonpositive. ☐

The linear shift $-p_H n$ is introduced so that the variational derivative of the energy becomes $p - p_H$. This makes the reaction contribution dissipative, since $n(p-p_H)G(p) \le 0$, and will be used in the Onsager formulation below.

3 Onsager formulation and numerical scheme

We now derive the numerical scheme from the Onsager variational principle. For a given density n, the instantaneous evolution is described by the density rate $\partial_t n$, the flux j, and the reaction rate r. To include transport and reaction in the same framework, we write the density equation as the balance law

$$\partial_t n + \nabla \cdot j = r. \tag{3.1}$$

The Onsager principle determines $\partial_t n$, j, and r by minimizing the Rayleighian

$$\mathcal{R} = \Phi + \frac{d}{dt}\mathcal{E}$$

under the constraint (3.1). Here Φ is a nonnegative quadratic dissipation functional, and $\frac{d}{dt}\mathcal{E}$ is the instantaneous variation of the modified energy $\mathcal{E}$ defined in (2.4). The Euler–Lagrange equations of this constrained minimization will recover the constitutive relations

$$j = -n\nabla p, \qquad r = nG(p).$$

Substituting these relations into (3.1), together with $p = n^\gamma$, gives the original model (1.1).

Choice of the Rayleighian. We now choose the dissipation functional Φ so that the above constrained minimization gives the desired constitutive relations. For notational simplicity, we introduce

$$\mu := \frac{\delta \mathcal{E}}{\delta n} = p - p_H. \tag{3.2}$$

In the continuous Onsager formulation, n is regarded as the current state and is not varied. Hence $p = n^\gamma$, μ, and the mobilities defined below are fixed coefficients in the minimization, while $\partial_t n$, j, and r are the variables.

For the transport part, we use the standard quadratic dissipation

$$\frac{1}{2}\int_\Omega \frac{|j|^2}{n}\,d\mathbf{x}.$$

When the Rayleighian is minimized subject to the balance law (3.1), this term leads to the flux relation

$$j = -n\nabla\mu.$$

Since $\nabla\mu = \nabla p$, the desired flux law $j = -n\nabla p$ is recovered.

The main point is the treatment of the reaction term. We use a quadratic reaction dissipation of the form

$$\frac{1}{2}\int_\Omega \frac{r^2}{\mathcal{M}(n,p)}\,d\mathbf{x}.$$

In the same Rayleighian minimization, this term leads to the Onsager relation

$$r = -\mathcal{M}(n,p)\mu.$$

To recover the growth term $r = nG(p)$, and using $\mu = p - p_H$, we first define the continuous extension of the secant slope

$$g(p) = \begin{cases} \dfrac{G(p)}{p - p_H}, & p \neq p_H, \\ G'(p_H), & p = p_H. \end{cases} \tag{3.3}$$

Then the reaction mobility is defined by

$$\mathcal{M}(n,p) = -ng(p). \tag{3.4}$$

By the assumptions on G (1.2)-(1.3), g is continuous and $g(p) \le 0$ for $p \ge 0$. Hence

$$\mathcal{M}(n,p) \ge 0 \qquad \text{for } n \ge 0, \quad p \ge 0.$$

Thus the reaction dissipation is nonnegative and yields the desired growth term through the Onsager relation.

Combining the transport and reaction parts, we define

$$\Phi(n; j, r) = \frac{1}{2}\int_\Omega \frac{|j|^2}{n}\, d\mathbf{x} + \frac{1}{2}\int_\Omega \frac{r^2}{\mathcal{M}(n,p)}\, d\mathbf{x}. \tag{3.5}$$

The dissipation terms in (3.5) should be understood in the extended sense. Namely, for $a \ge 0$,

$$\frac{|z|^2}{a} := \begin{cases} |z|^2/a, & a > 0, \\ 0, & a = 0,\ z = 0, \\ +\infty, & a = 0,\ z \ne 0. \end{cases}$$

With these notations, the corresponding Rayleighian is

$$\mathcal{R}(n; \partial_t n, j, r) = \Phi(n; j, r) + \int_\Omega \mu\, \partial_t n\, d\mathbf{x}. \tag{3.6}$$

Here the semicolon emphasizes that n is the given state, while $\partial_t n$, j, and r are the variables in the minimization.

Consistency verification. For completeness, we indicate the Euler–Lagrange relations. Introducing a Lagrange multiplier λ for the constraint (3.1), we define the augmented Rayleighian

$$\mathcal{R}_\lambda = \mathcal{R} - \int_\Omega \lambda\, (\partial_t n + \nabla\cdot j - r)\, d\mathbf{x}.$$

Taking the first variations of $\mathcal{R}_\lambda$ with respect to $\partial_t n$, j, and r gives

$$\lambda = \mu, \qquad \frac{j}{n} + \nabla\lambda = 0, \qquad \frac{r}{\mathcal{M}(n,p)} + \lambda = 0.$$

Hence

$$j = -n\nabla\mu, \qquad r = -\mathcal{M}(n,p)\mu. \tag{3.7}$$

Using the definition of μ (3.2) and $\mathcal{M}(n,p)$ (3.3)-(3.4), we recover

$$j = -n\nabla p, \qquad r = nG(p).$$

Thus the Onsager formulation is consistent with (1.1).

3.1 Time-discrete scheme from a discrete Rayleighian

We next derive a time-discrete scheme from the Rayleighian formulation and record its bound-preserving and energy-dissipating properties.

Let $\tau > 0$ be the time step and $t_k = k\tau$. Suppose that n^k is known and set

$$p^k = (n^k)^\gamma.$$

The mobilities in the dissipation terms are evaluated explicitly at the known state. Thus the transport mobility is n^k, and the reaction mobility is

$$\mathcal{M}^k = -n^k g(p^k). \tag{3.8}$$

By $g(p^k) \le 0$ for $p^k \ge 0$,

$$\mathcal{M}^k \ge 0 \qquad \text{if } n^k \ge 0. \tag{3.9}$$

We denote by m and q the time-integrated flux and reaction over one time step. The discrete balance law of (3.1) is

$$n - n^k + \nabla \cdot m = q, \qquad m \cdot \nu = 0 \quad \text{on } \partial\Omega. \tag{3.10}$$

Here the trial density n represents the new-time value, $m = \tau j$ and $q = \tau r$. Motivated by the continuous Rayleighian, we define the discrete Rayleighian

$$\mathcal{J}_\tau^k(n, m, q) = \mathcal{E}(n) - \mathcal{E}(n^k) + \frac{1}{2\tau} \int_\Omega \frac{|m|^2}{n^k} \, d\mathbf{x} + \frac{1}{2\tau} \int_\Omega \frac{q^2}{\mathcal{M}^k} \, d\mathbf{x}. \tag{3.11}$$

This functional is obtained by replacing $d\mathcal{E}/dt$ with the energy increment over one time step, freezing the mobilities at n^k.

The next time step is defined by the constrained minimization

$$(n^{k+1}, m^{k+1}, q^{k+1}) = \underset{(n,m,q) \in \mathcal{A}^k}{\arg\min} \; \mathcal{J}_\tau^k(n, m, q), \tag{3.12}$$

where

$$\mathcal{A}^k = \left\{ (n, m, q) \;\middle|\; n \ge 0, \quad n - n^k + \nabla \cdot m = q \text{ in } \Omega, \quad m \cdot \nu = 0 \text{ on } \partial\Omega \right\}. \tag{3.13}$$

Since the Rayleighian (3.11) is a sum of separate convex terms in n, m, and q, it is convex in the joint variable (n, m, q). Together with the convexity of the admissible set $\mathcal{A}^k$, this implies that the minimization problem (3.12) is convex.

The minimizer of (3.12) can be characterized by its Euler–Lagrange equations. Introducing a Lagrange multiplier λ for the discrete balance constraint (3.10), we define

$$\mathcal{J}_{\tau,\lambda}^k = \mathcal{J}_\tau^k - \int_\Omega \lambda \left(n - n^k + \nabla \cdot m - q \right) dx.$$

Let $(n^{k+1}, m^{k+1}, q^{k+1})$ be a smooth minimizer, and let λ^{k+1} be the corresponding multiplier. On the positivity set of n^{k+1}, taking variations with respect to n, m, q, and λ, we obtain

$$\frac{\delta \mathcal{E}}{\delta n}(n^{k+1}) = \lambda^{k+1}, \tag{3.14}$$
$$m^{k+1} = -\tau n^k \nabla \lambda^{k+1}, \tag{3.15}$$
$$q^{k+1} = -\tau \mathcal{M}^k \lambda^{k+1}, \tag{3.16}$$
$$n^{k+1} - n^k + \nabla \cdot m^{k+1} = q^{k+1}. \tag{3.17}$$

Here the boundary term in the variation with respect to m vanishes because of the boundary condition in (3.10).

Since

$$\lambda^{k+1} = \mu^{k+1} := p^{k+1} - p_H, \qquad p^{k+1} = (n^{k+1})^\gamma,$$

substituting (3.15)–(3.16) into (3.17) gives

$$\frac{n^{k+1} - n^k}{\tau} = \nabla \cdot \left(n^k \nabla \mu^{k+1} \right) - \mathcal{M}^k \mu^{k+1}. \tag{3.18}$$

Equivalently, using $\mu^{k+1} = p^{k+1} - p_H$, we obtain the pressure form

$$\frac{n^{k+1} - n^k}{\tau} = \nabla \cdot \left(n^k \nabla p^{k+1} \right) - \mathcal{M}^k \left(p^{k+1} - p_H \right), \qquad p^{k+1} = (n^{k+1})^\gamma. \tag{3.19}$$

Moreover, from $m^{k+1} \cdot \nu = 0$ on $\partial\Omega$, we recover the boundary condition

$$n^k \nabla p^{k+1} \cdot \nu = 0 \qquad \text{on } \partial\Omega. \tag{3.20}$$

The resulting scheme can be viewed as a natural time-discrete counterpart of (1.1). Its main feature is the explicit evaluation of the mobilities together with the implicit update of the pressure. This explicit–implicit treatment keeps the scheme simple while preserving the main structural properties of the continuous model.

Since the minimization problem (3.12) is convex, the Euler–Lagrange system (3.19) derived above is not only a necessary condition for a smooth minimizer, but also a sufficient condition for global minimality, whenever the corresponding solution is admissible. In this sense, the time-discrete scheme may be equivalently described by the constrained minimization problem (3.12) or by its Euler–Lagrange equations (3.19).

Before passing to the fully discrete finite difference formulation, we show the positivity, homeostatic upper bound, and energy dissipation of the time-discrete scheme.

Proposition 3.1 (Positivity and homeostatic upper bound)**.** *Assume that $n^k \geq 0$. Let $(n^{k+1}, m^{k+1}, q^{k+1})$ be a minimizer of (3.12). Then*

$$n^{k+1} \geq 0.$$

If, in addition, $n^k \leq n_H := p_H^{\frac{1}{\gamma}}$ and the minimizer satisfies (3.19)–(3.20), then

$$0 \leq n^{k+1} \leq n_H, \qquad 0 \leq p^{k+1} \leq p_H.$$

Proof. The nonnegativity follows directly from the admissible set $\mathcal{A}^k$, where $n \geq 0$ is imposed.

To prove the upper bound, set

$$w = (p^{k+1} - p_H)_+.$$

On the set $\{w > 0\}$, we have $p^{k+1} > p_H$, hence $n^{k+1} > n_H \geq n^k$. Therefore

$$(n^{k+1} - n^k)w > 0.$$

Multiplying (3.19) by w, integrating over Ω, and using (3.20), we obtain

$$\frac{1}{\tau}\int_\Omega (n^{k+1} - n^k) w \, dx = -\int_\Omega n^k |\nabla w|^2 \, dx - \int_\Omega \mathcal{M}^k w^2 \, dx \leq 0.$$

The left-hand side is nonnegative, and hence it must vanish. We conclude that $w = 0$ and thus $p^{k+1} \leq p_H$. Together with $n^{k+1} \geq 0$ and $p^{k+1} = (n^{k+1})^\gamma$, this gives

$$0 \leq p^{k+1} \leq p_H, \qquad 0 \leq n^{k+1} \leq n_H.$$

□

Proposition 3.2 (Energy stability)**.** *Assume that $n^k \geq 0$. Let $(n^{k+1}, m^{k+1}, q^{k+1})$ be a minimizer of (3.12). Then*

$$\mathcal{E}(n^{k+1}) + \frac{1}{2\tau}\int_\Omega \frac{|m^{k+1}|^2}{n^k} \, dx + \frac{1}{2\tau}\int_\Omega \frac{|q^{k+1}|^2}{\mathcal{M}^k} \, dx \leq \mathcal{E}(n^k). \tag{3.21}$$

In particular, $\mathcal{E}(n^{k+1}) \leq \mathcal{E}(n^k)$.

Proof. Since $(n^k, 0, 0) \in \mathcal{A}^k$, the minimizing property gives

$$\mathcal{J}_\tau^k(n^{k+1}, m^{k+1}, q^{k+1}) \leq \mathcal{J}_\tau^k(n^k, 0, 0) = 0.$$

Using the definition of $\mathcal{J}_\tau^k$ gives (3.21). The last assertion follows from the nonnegativity of the two dissipation terms. □

3.2 Fully discrete finite difference scheme

We now present the fully discrete finite difference scheme. For simplicity, we give the one-dimensional formulation on $\Omega = (a, b)$. The two-dimensional extension used in the computations is stated in Appendix A.

Let $h = (b - a)/N$, $x_{i+\frac{1}{2}} = a + ih$ for $0 \le i \le N$, and $x_i = a + (i - \frac{1}{2})h$ for $1 \le i \le N$. We denote by n_i^k the cell-centered approximation of $n(x_i, t_k)$, and set $p_i^k = (n_i^k)^\gamma$. For a cell-centered grid function $v = \{v_i\}_{i=1}^N$ and an interface grid function $F = \{F_{i+\frac{1}{2}}\}_{i=0}^N$, define

$$(\delta_h v)_{i+\frac{1}{2}} = \frac{v_{i+1} - v_i}{h}, \qquad (d_h F)_i = \frac{F_{i+\frac{1}{2}} - F_{i-\frac{1}{2}}}{h}. \tag{3.22}$$

Given n^k, the new density n^{k+1} is determined as follows. Let

$$n_{i+\frac{1}{2}}^k = \frac{n_i^k + n_{i+1}^k}{2}, \quad F_{i+\frac{1}{2}}^{k+1} = n_{i+\frac{1}{2}}^k (\delta_h p^{k+1})_{i+\frac{1}{2}}, \quad 1 \le i \le N-1. \tag{3.23}$$

Here F denotes the pressure-gradient flux $n\nabla p$, that is, the negative of the physical flux j. The no-flux boundary condition is imposed by

$$F_{\frac{1}{2}}^{k+1} = F_{N+\frac{1}{2}}^{k+1} = 0. \tag{3.24}$$

Then the fully discrete scheme is

$$\frac{n_i^{k+1} - n_i^k}{\tau} = (d_h F^{k+1})_i - \mathcal{M}_i^k \left(p_i^{k+1} - p_H \right), \qquad 1 \le i \le N, \tag{3.25}$$

where

$$\mathcal{M}_i^k = -n_i^k g(p_i^k). \tag{3.26}$$

Here $p_i^{k+1} = (n_i^{k+1})^\gamma$. Since $g(p_i^k) \le 0$ for $p_i^k \ge 0$, if $n_i^k \ge 0$, then

$$\mathcal{M}_i^k \ge 0, \qquad 1 \le i \le N. \tag{3.27}$$

The fully discrete scheme (3.25) is a nonlinear system for n^{k+1}, which can be solved by a damped Newton iteration in the computations. See Appendix B. Below is the fully discrete update of n^k in one time step.

Fully discrete update. Given n^k,

1. Set $p_i^k = (n_i^k)^\gamma$.
2. Compute $\mathcal{M}_i^k = -n_i^k g(p_i^k)$.
3. Solve the nonlinear system (3.25) for n_i^{k+1}, with $p_i^{k+1} = (n_i^{k+1})^\gamma$.

Proposition 3.3 (Existence and nonnegativity)**.** *Assume that $n_i^k \ge 0$ for $1 \le i \le N$. Then the fully discrete scheme (3.23)–(3.26) admits a nonnegative solution*

$$n_i^{k+1} \ge 0, \qquad 1 \le i \le N.$$

Proof. We use the pressure variable. For $p = \{p_i\}_{i=1}^N$, consider the functional

$$\mathcal{J}_h(p) = h \sum_{i=1}^N \left[\frac{\gamma}{\gamma+1} p_i^{\frac{\gamma+1}{\gamma}} - n_i^k p_i + \frac{\tau}{2} \mathcal{M}_i^k (p_i - p_H)^2 \right] + \frac{\tau h}{2} \sum_{i=1}^{N-1} n_{i+\frac{1}{2}}^k \left| (\delta_h p)_{i+\frac{1}{2}} \right|^2$$

on the closed convex set

$$K_h = \{p \in \mathbb{R}^N : \ p_i \geq 0, \ 1 \leq i \leq N\}.$$

Since $n^k_{i+\frac{1}{2}} \geq 0$ and $\mathcal{M}^k_i \geq 0$, the functional $\mathcal{J}_h$ is convex. Moreover, the term $p_i^{(\gamma+1)/\gamma}$ is superlinear on $[0,\infty)$, so $\mathcal{J}_h$ is coercive on K_h. Hence $\mathcal{J}_h$ admits a minimizer $p^{k+1} \in K_h$.

We now show that this minimizer satisfies the finite difference scheme. Let

$$R_i(p) = \frac{1}{h}\frac{\partial \mathcal{J}_h}{\partial p_i} = p_i^{1/\gamma} - n^k_i + \tau \mathcal{M}^k_i (p_i - p_H) - \tau (d_h F)_i.$$

The variational inequality for the minimizer gives $R_i(p^{k+1}) = 0$ if $p^{k+1}_i > 0$, and $R_i(p^{k+1}) \geq 0$ if $p^{k+1}_i = 0$. In the latter case, p^{k+1}_i is a minimum of the nonnegative grid function p^{k+1}, and hence $(d_h F)_i \geq 0$. Therefore

$$R_i(p^{k+1}) = -n^k_i - \tau \mathcal{M}^k_i p_H - \tau (d_h F)_i \leq 0.$$

Thus $R_i(p^{k+1}) = 0$ for all i. Setting $n^{k+1}_i = (p^{k+1}_i)^{1/\gamma}$, we obtain

$$\frac{n^{k+1}_i - n^k_i}{\tau} = (d_h F^{k+1})_i - \mathcal{M}^k_i (p^{k+1}_i - p_H),$$

which is exactly (3.25). Since $p^{k+1} \in K_h$, the solution is nonnegative. □

Proposition 3.4 (Homeostatic upper bound)**.** *Assume that*

$$0 \leq n^k_i \leq n_H := p_H^{1/\gamma}, \qquad 1 \leq i \leq N.$$

Let n^{k+1} be a nonnegative solution of (3.23)–(3.26). Then

$$0 \leq n^{k+1}_i \leq n_H, \qquad 0 \leq p^{k+1}_i \leq p_H, \qquad 1 \leq i \leq N.$$

Proof. Assume by contradiction that $\max_i p^{k+1}_i > p_H$, and let ℓ be such that $p^{k+1}_\ell = \max_i p^{k+1}_i$. Then $n^{k+1}_\ell > n_H$ and $p^{k+1}_\ell > p_H$. Since p^{k+1}_ℓ is a maximum and $n^k_{i+\frac{1}{2}} \geq 0$, we have

$$(d_h F^{k+1})_\ell \leq 0.$$

Using the scheme at $i = \ell$, we get

$$\frac{n^{k+1}_\ell - n^k_\ell}{\tau} = (d_h F^{k+1})_\ell - \mathcal{M}^k_\ell (p^{k+1}_\ell - p_H) \leq 0.$$

Hence $n^{k+1}_\ell \leq n^k_\ell \leq n_H$, contradicting $n^{k+1}_\ell > n_H$. Therefore $n^{k+1}_i \leq n_H$ for all i. Together with nonnegativity and $p^{k+1}_i = (n^{k+1}_i)^\gamma$, this gives the desired bounds. □

Proposition 3.5 (Discrete energy dissipation)**.** *Assume that $n^k_i \geq 0$ for $1 \leq i \leq N$. Define*

$$\mathcal{E}_h(n) = h \sum_{i=1}^{N} \left(\frac{1}{\gamma+1} n_i^{\gamma+1} - p_H n_i \right). \tag{3.28}$$

Let n^{k+1} be a nonnegative solution of (3.23)–(3.26). Then

$$\mathcal{E}_h(n^{k+1}) \leq \mathcal{E}_h(n^k). \tag{3.29}$$

Proof. Let

$$e(n) = \frac{1}{\gamma+1} n^{\gamma+1} - p_H n.$$

Since e is convex and $e'(n) = n^\gamma - p_H$, we have

$$e(n_i^{k+1}) - e(n_i^k) \le (p_i^{k+1} - p_H)(n_i^{k+1} - n_i^k).$$

Multiplying (3.25) by $\tau h(p_i^{k+1} - p_H)$, summing over i, and using the above inequality, we obtain

$$\mathcal{E}_h(n^{k+1}) - \mathcal{E}_h(n^k) \le \tau h \sum_{i=1}^{N} (p_i^{k+1} - p_H)(d_h F^{k+1})_i - \tau h \sum_{i=1}^{N} \mathcal{M}_i^k |p_i^{k+1} - p_H|^2. \tag{3.30}$$

By summation by parts and the no-flux condition (3.24),

$$h \sum_{i=1}^{N} (p_i^{k+1} - p_H)(d_h F^{k+1})_i = -h \sum_{i=1}^{N-1} F_{i+\frac{1}{2}}^{k+1} (\delta_h p^{k+1})_{i+\frac{1}{2}} = -h \sum_{i=1}^{N-1} n_{i+\frac{1}{2}}^k \left|(\delta_h p^{k+1})_{i+\frac{1}{2}}\right|^2. \tag{3.31}$$

Here the last equality follows from the definition of $F_{i+\frac{1}{2}}^{k+1}$ in (3.23). Combining (3.30) and (3.31) yields

$$\mathcal{E}_h(n^{k+1}) + \tau h \sum_{i=1}^{N-1} n_{i+\frac{1}{2}}^k \left|(\delta_h p^{k+1})_{i+\frac{1}{2}}\right|^2 + \tau h \sum_{i=1}^{N} \mathcal{M}_i^k \left|p_i^{k+1} - p_H\right|^2 \le \mathcal{E}_h(n^k). \tag{3.32}$$

Dropping the nonnegative dissipation terms yields $\mathcal{E}_h(n^{k+1}) \le \mathcal{E}_h(n^k)$. □

Combining Propositions 3.3, 3.4, and 3.5, we immediately get the following corollary.

Corollary 3.1 (Structure-preserving properties). *If $0 \le n_i^0 \le n_H$ for $1 \le i \le N$, then for all k,*

$$0 \le n_i^k \le n_H, \qquad 0 \le p_i^k \le p_H,$$

and $\mathcal{E}_h(n^k)$ (3.28) is nonincreasing.

3.3 Fixed-grid stiff-pressure limiting structure

We now analyze the fully discrete scheme in the stiff-pressure regime $\gamma \to \infty$. The mesh size h and the time step τ are kept fixed throughout this subsection.

To emphasize the dependence on γ, in this subsection we write

$$n_{\gamma,i}^k, \qquad p_{\gamma,i}^k = (n_{\gamma,i}^k)^\gamma$$

for the solution of the fully discrete scheme (3.25)–(3.26). The corresponding numerical flux and reaction mobility are denoted by $F_{\gamma,i+\frac{1}{2}}^{k+1}$ and $\mathcal{M}_{\gamma,i}^k$, respectively. They are defined by the same formulas as in (3.23) and (3.26), with n_i^k, p_i^k replaced by $n_{\gamma,i}^k, p_{\gamma,i}^k$. The no-flux boundary condition is imposed as before.

Proposition 3.6 (Fixed-grid stiff-pressure limiting structure). *Assume that G satisfies (1.2)-(1.3). Let h, τ, and the total time steps $K \ge 1$ be fixed. Assume that, for any $\gamma > 1$, the initial data satisfy*

$$0 \le p_{\gamma,i}^0 \le p_H, \qquad 1 \le i \le N,$$

and that, up to a subsequence,

$$n_{\gamma,i}^0 \to n_i^0, \qquad p_{\gamma,i}^0 \to p_i^0, \qquad 1 \le i \le N.$$

Then there exists a further subsequence, still denoted by γ, such that, for all $0 \le k \le K$,

$$n_{\gamma,i}^k \to n_i^k, \qquad p_{\gamma,i}^k \to p_i^k, \qquad 1 \le i \le N.$$

For every $0 \le k \le K$, the limiting variables satisfy the discrete counterpart of (1.4), i.e.

$$0 \le n_i^k \le 1, \qquad 0 \le p_i^k \le p_H, \qquad p_i^k(1 - n_i^k) = 0. \tag{3.33}$$

Moreover, for $0 \le k \le K-1$, they satisfy the limiting discrete density equation

$$\frac{n_i^{k+1} - n_i^k}{\tau} = (d_h F^{k+1})_i - \mathcal{M}_i^k \left(p_i^{k+1} - p_H\right), \qquad 1 \le i \le N, \tag{3.34}$$

where F^{k+1} and $\mathcal{M}^k$ are defined by the same formulas as (3.23) and (3.26), with $(n^k, p^k, n^{k+1}, p^{k+1})$ replaced by the limiting variables.

Proof. The bound-preserving property gives, uniformly in γ,

$$0 \le n_{\gamma,i}^k \le p_H^{1/\gamma}, \qquad 0 \le p_{\gamma,i}^k \le p_H, \qquad 1 \le i \le N, \quad 0 \le k \le K.$$

Since h, τ, N, and K are fixed, the grid-time space is finite dimensional. Hence, up to a subsequence,

$$n_{\gamma,i}^k \to n_i^k, \qquad p_{\gamma,i}^k \to p_i^k, \qquad 1 \le i \le N, \quad 0 \le k \le K.$$

Passing to the limit in the bounds gives

$$0 \le n_i^k \le 1, \qquad 0 \le p_i^k \le p_H.$$

The complementarity relation follows directly from the pressure law. Since $p_{\gamma,i}^k = (n_{\gamma,i}^k)^\gamma$, we have

$$p_{\gamma,i}^k(1 - n_{\gamma,i}^k) = p_{\gamma,i}^k \left(1 - (p_{\gamma,i}^k)^{1/\gamma}\right).$$

Using the uniform bound $0 \le p_{\gamma,i}^k \le C$, one obtains

$$\sup_{0 \le a \le C} \left| a(1 - a^{1/\gamma}) \right| \to 0, \qquad \gamma \to \infty.$$

Hence

$$p_{\gamma,i}^k(1 - n_{\gamma,i}^k) \to 0, \quad \gamma \to \infty.$$

Passing to the limit gives

$$p_i^k(1 - n_i^k) = 0.$$

It remains to pass to the limit in the scheme. Since the difference operators are finite-dimensional linear operators, we have

$$F_{\gamma,i+\frac{1}{2}}^{k+1} \to n_{i+\frac{1}{2}}^k (\delta_h p^{k+1})_{i+\frac{1}{2}} = F_{i+\frac{1}{2}}^{k+1}.$$

For the mobility, using the continuous function g defined in (3.3), we have

$$\mathcal{M}_{\gamma,i}^k = -n_{\gamma,i}^k g(p_{\gamma,i}^k) \to -n_i^k g(p_i^k) = \mathcal{M}_i^k.$$

Passing to the limit in the fully discrete scheme gives (3.34). □

Consistency with the Hele–Shaw limit (1.6a)**.** We next rewrite the limiting scheme in a discrete pressure-complementarity form. Define the linearly implicit approximation of the growth rate by

$$G_{i,\mathrm{lin}}^k(s) := g(p_i^k)(s - p_H). \tag{3.35}$$

This is the secant approximation of $G(\cdot)$ anchored at p_H, with the tangent value used at $p_i^k = p_H$. It satisfies

$$G_{i,\mathrm{lin}}^k(p_H) = 0, \qquad G_{i,\mathrm{lin}}^k(p_i^k) = G(p_i^k), \qquad (s - p_H)G_{i,\mathrm{lin}}^k(s) \le 0.$$

With the notation in (3.35), the reaction source in the limiting scheme is written as $n_i^k G_{i,\mathrm{lin}}^k(s)$. Indeed, by (3.26), we have

$$-\mathcal{M}_i^k(s - p_H) = n_i^k G_{i,\mathrm{lin}}^k(s).$$

Then the limiting discrete density equation (3.34) becomes

$$\frac{n_i^{k+1} - n_i^k}{\tau} = (d_h F^{k+1})_i + n_i^k G_{i,\mathrm{lin}}^k(p_i^{k+1}). \tag{3.36}$$

Combining (3.36) with the complementarity relation $p_i^{k+1}(1 - n_i^{k+1}) = 0$ in (3.33) gives

$$p_i^{k+1}\left[\frac{1 - n_i^k}{\tau} - (d_h F^{k+1})_i - n_i^k G_{i,\mathrm{lin}}^k(p_i^{k+1})\right] = 0. \tag{3.37}$$

To identify the discrete pressure equation in saturated cells, set $\eta_{i+\frac{1}{2}}^k := 1 - n_{i+\frac{1}{2}}^k$. Since

$$(d_h F^{k+1})_i = (\delta_h^2 p^{k+1})_i - \big(d_h(\eta^k \delta_h p^{k+1})\big)_i,$$

we can rewrite (3.37) as

$$p_i^{k+1}\left[\frac{1 - n_i^k}{\tau} - (\delta_h^2 p^{k+1})_i + \big(d_h(\eta^k \delta_h p^{k+1})\big)_i - n_i^k G_{i,\mathrm{lin}}^k(p_i^{k+1})\right] = 0.$$

If $p_i^{k+1} > 0$ and the neighboring old-time cells are saturated, namely $n_{i-1}^k = n_i^k = n_{i+1}^k = 1$, then $1 - n_i^k = 0$ and $\eta_{i-\frac{1}{2}}^k = \eta_{i+\frac{1}{2}}^k = 0$. Hence the above relation reduces to

$$(\delta_h^2 p^{k+1})_i + G_{i,\mathrm{lin}}^k(p_i^{k+1}) = 0. \tag{3.38}$$

This is the finite difference Hele–Shaw-type pressure equation obtained from the fixed-grid stiff-pressure limit. If G is linear, then $G_{i,\mathrm{lin}}^k(s) = G(s)$, and (3.38) becomes the standard finite difference form of

$$\Delta p + G(p) = 0 \qquad \text{in } \{p > 0\}.$$

For nonlinear G, $G_{i,\mathrm{lin}}^k$ is the secant linearization anchored at p_H, with its tangent limit used when $p_i^k = p_H$.

Consistency with the interface motion (1.6c)**.** We finally discuss how the limiting difference equation reflects the free-boundary motion. For simplicity, consider a one-dimensional symmetric patch on the interval $(0, b)$, with symmetry at $x = 0$ and with the right free boundary $R(t) < b$. In the Hele–Shaw limit, the density has the patch form

$$n(x,t) = \begin{cases} 1, & 0 < x < R(t), \\ 0, & R(t) < x < b, \end{cases} \tag{3.39}$$

and the pressure satisfies

$$p(x,t) > 0 \quad \text{for } 0 < x < R(t), \qquad p(x,t) = 0 \quad \text{for } R(t) < x < b.$$

The pressure is continuous across the free boundary, so $p(R(t),t)=0$. The left endpoint is the symmetry point, and hence $p_x(0,t)=0$.

In this symmetric setting, the occupied length on $(0,b)$ is exactly $R(t)$. Therefore the right boundary velocity can be obtained from the time variation of the total mass

$$R'(t) = \frac{d}{dt}\int_0^b n(x,t)\,dx.$$

The same idea has a direct discrete analogue. Define the discrete total mass on $(0,b)$ by

$$m_h^k := h\sum_{j=1}^N n_j^k.$$

In the stiff-pressure limit, m_h^k approximates the occupied length. Hence the mesh-scale right boundary velocity is naturally defined by

$$V_h^{k+1} := \frac{m_h^{k+1}-m_h^k}{\tau} = h\sum_{j=1}^N \frac{n_j^{k+1}-n_j^k}{\tau}. \tag{3.40}$$

Summing (3.36) over all grid indices and using the definition (3.40), we obtain

$$V_h^{k+1} = h\sum_{j=1}^N (d_h F^{k+1})_j + h\sum_{j=1}^N n_j^k G_{j,\mathrm{lin}}^k\big(p_j^{k+1}\big). \tag{3.41}$$

Using the definition of d_h in (3.22), we have

$$h\sum_{j=1}^N (d_h F^{k+1})_j = F^{k+1}_{N+\frac{1}{2}} - F^{k+1}_{\frac{1}{2}}.$$

The left boundary is the symmetry point, and the right computational boundary is taken far enough with no flux. Therefore

$$F^{k+1}_{\frac{1}{2}} = F^{k+1}_{N+\frac{1}{2}} = 0.$$

Thus (3.41) reduces to the global discrete relation

$$V_h^{k+1} = h\sum_{j=1}^N n_j^k G_{j,\mathrm{lin}}^k\big(p_j^{k+1}\big). \tag{3.42}$$

We compare (3.42) with the continuous Hele–Shaw velocity law (1.6c). In the continuous limit, the pressure satisfies (1.6a), the Neumann boundary condition at $x=0$ and the Dirichlet boundary condition (1.6b) at the free boundary $x=R(t)$. That is,

$$-p_{xx} = G(p), \qquad 0<x<R(t), \qquad p_x(0,t)=0, \qquad p(R(t),t)=0.$$

Integrating over $(0,R(t))$ gives

$$-p_x(R(t),t) + p_x(0,t) = \int_0^{R(t)} G\big(p(x,t)\big)\,dx.$$

Using $p_x(0,t)=0$ and the velocity law in (1.6c), we obtain

$$R'(t) = -p_x(R(t),t) = \int_0^{R(t)} G\big(p(x,t)\big)\,dx = \int_0^b n(x,t)G\big(p(x,t)\big)\,dx. \tag{3.43}$$

Here the last equality follows from the patch structure (3.39). Therefore the global discrete relation (3.42) is consistent with the continuous Hele–Shaw interface identity (3.43) in this setting.

4 Numerical experiments

In this section, we test the proposed fully discrete scheme. We first test the degenerate diffusion part and the accuracy of the growth model. We then focus on the fixed-grid stiff-pressure limit $\gamma \to \infty$, and finally present two-dimensional examples with topology changes.

Unless otherwise stated, homogeneous no-flux boundary conditions are imposed. We use $G(p) = 0$ for the pure diffusion Barenblatt benchmark, and choose

$$G(p) = 1 - p, \qquad p_H = 1,$$

for all tests involving pressure-dependent growth. This choice satisfies (1.2) and (1.3), and the corresponding homeostatic density is $n_H = 1$.

4.1 Accuracy and structure-preserving tests

Barenblatt benchmark for degenerate diffusion. When $G(p) = 0$, the equation (1.1) in 1D becomes the porous medium equation

$$\partial_t n = \partial_x \left(n \partial_x n^\gamma \right) = \frac{\gamma}{\gamma + 1} \partial_{xx} n^{\gamma+1}. \tag{4.1}$$

This benchmark checks the degenerate diffusion part of the scheme, including compact support, mass conservation, and energy dissipation.

Equation (4.1) admits a time-shifted Barenblatt solution. With $\kappa = \gamma/(\gamma+1)$, $\beta = 1/(\gamma+2)$, and $s = \kappa(t + t_0)$, it is given by

$$n_{\text{ex}}(x,t) = s^{-\beta} \left(C - \frac{\beta\gamma}{2(\gamma+1)} \frac{x^2}{s^{2\beta}} \right)_+^{1/\gamma}, \qquad (z)_+ = \max\{z, 0\}. \tag{4.2}$$

We take $n_0(x) = n_{\text{ex}}(x, 0)$. For the profile comparison, we use $\Omega = [-5, 5]$, $h = 1/64$, $\tau = 0.01$, $T = 1$, and $t_0 = 2$. Two values of γ are tested: $C = 1$ for $\gamma = 3$, and $C = 0.1$ for $\gamma = 20$, so that the support remains away from the boundary. Figure 1 shows that the numerical solution agrees well with the Barenblatt profile at $T = 1$.

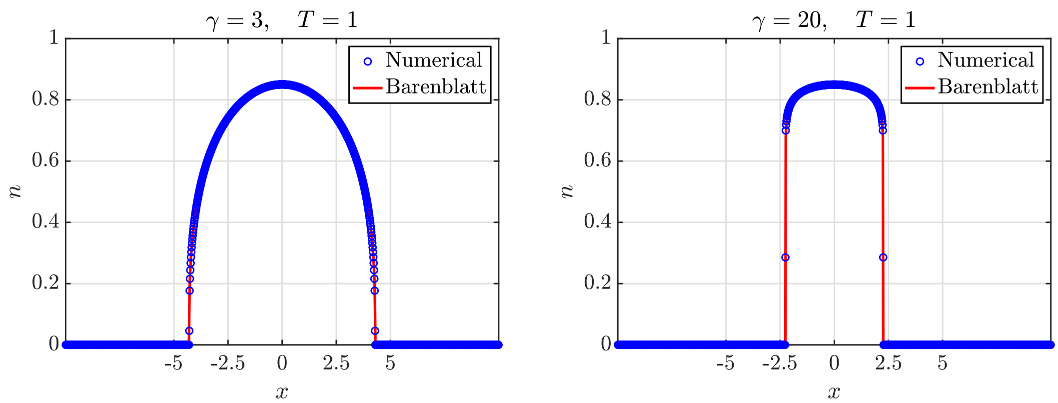

Figure 1: Comparison between the numerical solution and the Barenblatt profile for the porous medium equation (4.1). Left: $\gamma = 3$, $C = 1$. Right: $\gamma = 20$, $C = 0.1$.

Since $G(p) = 0$, the total mass is conserved and the reaction mobility vanishes. The linear shift in the modified energy is therefore irrelevant, and we plot the standard porous-medium energy

$$E_h(t^k) = h \sum_{i=1}^{N} \frac{(n_i^k)^{\gamma+1}}{\gamma + 1}.$$

We define $m_h(t^k) := h\sum_{i=1}^{N} n_i^k$, and monitor the relative mass variation and the normalized energy,

$$\frac{m_h(t^k)-m_h(0)}{m_h(0)}, \qquad \frac{E_h(t^k)}{E_h(0)}.$$

For the long-time conservation and dissipation diagnostics, we extend the computational interval to $[-10, 10]$ and the final time to $T = 100$, so that the support remains away from the boundary. Figure 2 shows that the mass variation stays at round-off level and that the normalized energy is nonincreasing.

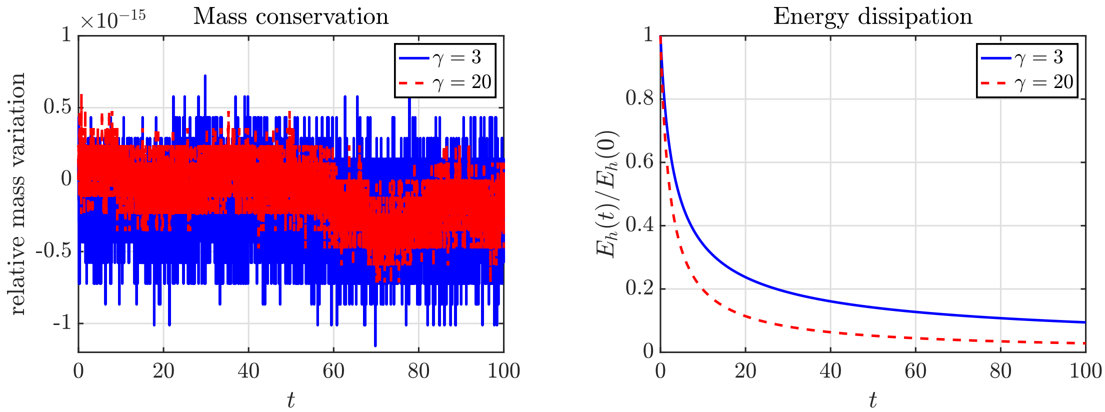

Figure 2: Mass conservation and energy dissipation for the Barenblatt test. Left: relative mass variation. Right: normalized energy $E_h(t)/E_h(0)$.

Accuracy test. We now test the convergence behavior of the fully discrete scheme using the same Barenblatt-type initial density as in the preceding benchmark. Both the pure diffusion case and the pressure-dependent growth case specified at the beginning of this section are considered.

Let $n_{h,\tau,i}^k$ be the numerical density computed with mesh size h and time step τ. The error reported in Figure 3 is defined by

$$\mathcal{E}_{h,\tau} = \max_k \left\{ h\sum_{i=1}^{N} \left| n_{h,\tau,i}^k - n_{\mathrm{ref}}(x_i, t^k) \right| \right\}.$$

Thus the error is measured in the discrete L^1 norm in space and then maximized over time.

In the spatial refinement test, n_{ref} is the analytical Barenblatt solution for $G(p) = 0$ and the finest-grid numerical solution, interpolated to the corresponding cell centers, for $G(p) = 1 - p$. In the temporal refinement test, n_{ref} is computed using the smallest time step on the same spatial grid, so that the reported error isolates the temporal discretization error on a fixed mesh rather than the total error to the exact solution. We fix $\tau = 10^{-5}$ for the spatial refinement and $N = 4096$ for the temporal refinement. The observed results in Figure 3 show a consistent decrease in the error under both refinements, with rates close to one.

Structure preservation for the growth model. We next check the discrete energy-dissipation property in the pressure-dependent growth case $G(p) = 1 - p$. Unlike the pure diffusion case, the total mass is not conserved because of proliferation. We therefore monitor the normalized mass $m_h(t)/m_h(0)$ and the modified discrete energy

$$E_h(t^k) = h\sum_{i=1}^{N} \left(\frac{(n_i^k)^{\gamma+1}}{\gamma+1} - p_H n_i^k \right).$$

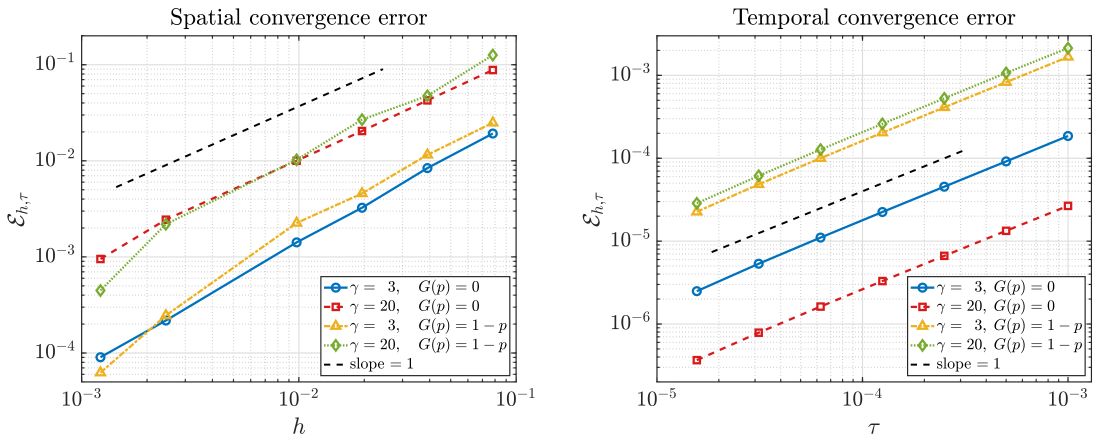

Figure 3: Convergence errors for the one-dimensional accuracy test. Left: spatial refinement with $\tau = 10^{-5}$. Right: temporal refinement with $N = 4096$.

The initial density is the same as in the preceding accuracy test. For this long-time diagnostic, we use $\Omega = [-20, 20]$, $h = 1/64$, $\tau = 0.01$, and run the computation up to $T = 10$. Figure 4 shows that the mass changes due to proliferation, while the modified energy remains nonincreasing, in agreement with the discrete dissipation estimate in Proposition 3.5.

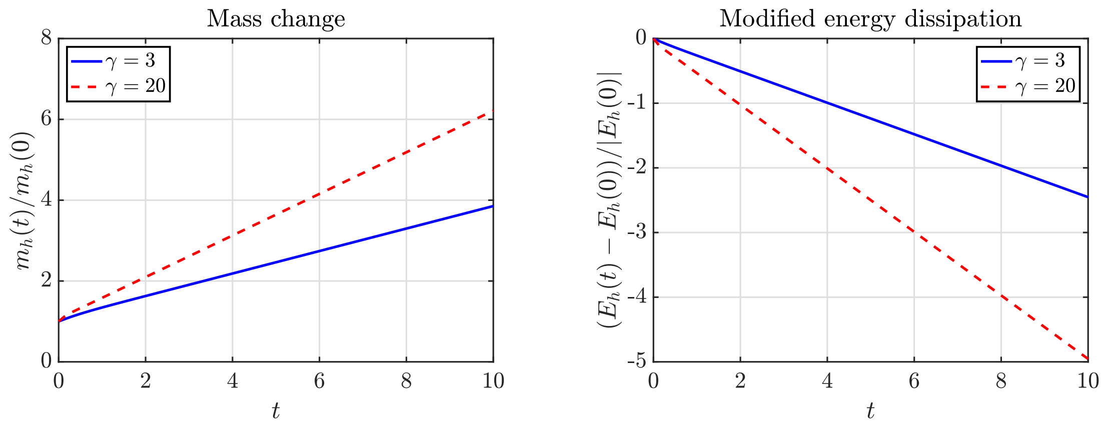

Figure 4: Mass change and modified energy dissipation for the growth model with $G(p) = 1 - p$. Left: normalized mass $m_h(t)/m_h(0)$. Right: normalized modified energy change $(E_h(t) - E_h(0))/|E_h(0)|$.

4.2 Fixed-grid stiff-pressure limit

Hele–Shaw reference solution and setup. We first describe the reference Hele–Shaw solution used in the stiff-pressure test. According to (1.6a)–(1.6b) with $G(p) = 1 - p$, a one-dimensional symmetric patch $(-R(t), R(t))$ satisfies

$$-p_{\infty,xx} = 1 - p_\infty, \quad |x| < R(t), \qquad p_\infty(\pm R(t), t) = 0.$$

By symmetry, $p_{\infty,x}(0, t) = 0$. Solving this boundary-value problem, together with the boundary velocity law in (1.6c), gives the explicit reference solution

$$p_\infty(x,t) = \left(1 - \frac{\cosh x}{\cosh R(t)}\right)_+, \quad n_\infty(x,t) = \mathbf{1}_{[-R(t),R(t)]}(x), \quad R(t) = \operatorname{arsinh}\big(e^t \sinh R_0\big). \tag{4.3}$$

For the finite-γ computations, with $R_0 = 1$, we take

$$p_0(x) = \left(1 - \frac{\cosh x}{\cosh R_0}\right)_+, \qquad n_{0,\gamma}(x) = p_0(x)^{1/\gamma}.$$

All computations in this subsection use $\Omega = [-5, 5]$, $N = 6400$, $h = 10/N$, $\tau = 10^{-4}$, and $T = 1$. The same h and τ are used for all values of γ, which gives a fixed-grid test of the stiff-pressure limit discussed in Section 3.3.

Convergence to the Hele–Shaw profile. For the profile comparison, we display the numerical solutions for $\gamma = 3, 20, 200$. Figure 5 shows the density and pressure at $T = 1$, together with the explicit Hele–Shaw profiles in (4.3). As γ increases, the numerical density approaches the characteristic function n_∞, while the pressure converges to the limiting Hele–Shaw pressure. The transition in the density becomes sharper near the free boundary, whereas the pressure profile remains smooth and continuous.

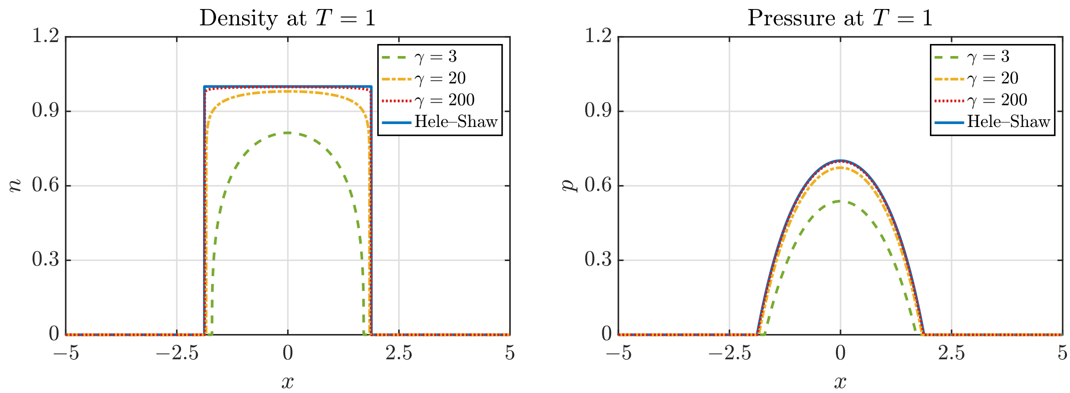

Figure 5: Fixed-grid stiff-pressure limit in one dimension at $T = 1$. Left: density profiles. Right: pressure profiles. The solid curves denote the explicit Hele–Shaw limit.

For the quantitative comparison, let $K = T/\tau$. We first measure the final-time L^1 differences in density and pressure by

$$e_n^\gamma(T) = h \sum_{i=1}^{N} \left| n_{\gamma,i}^K - n_\infty(x_i, T) \right|, \qquad e_p^\gamma(T) = h \sum_{i=1}^{N} \left| p_{\gamma,i}^K - p_\infty(x_i, T) \right|.$$

For the symmetric patch limit, we also define the mass-based effective radius and its error by

$$R_{\gamma,h}(T) = \frac{1}{2} h \sum_{i=1}^{N} n_{\gamma,i}^K, \qquad e_R^\gamma(T) = |R_{\gamma,h}(T) - R(T)|.$$

Motivated by the discrete complementarity relation (3.33), we compute

$$\mathcal{C}_\gamma(T) = h \sum_{i=1}^{N} \left| p_{\gamma,i}^K (1 - n_{\gamma,i}^K) \right|.$$

To check the interior discrete Hele–Shaw pressure equation (3.38), we evaluate the residual away from the interface on

$$\mathcal{I}_\eta(T) = \{ i : \; p_\infty(x_i, T) \geq \eta \}, \qquad \eta = 0.2,$$

and define

$$\mathcal{H}_{\gamma,\eta}(T) = h \sum_{i \in \mathcal{I}_\eta(T)} \left| (\delta_h^2 p_\gamma^K)_i + 1 - p_{\gamma,i}^K \right|.$$

Here δ_h^2 denotes the centered second difference. The final-time diagnostics in Table 1 show that all errors decrease as γ increases, confirming the convergence toward the analytical Hele–Shaw limit and the discrete complementarity structure. The observed orders are close to one, suggesting an approximately first-order decay with respect to $1/\gamma$.

Table 1: Final-time diagnostics for the fixed-grid stiff-pressure test with $N = 6400$, $T = 1$ and $\eta = 0.2$.

γ	10	20	40	80	160	320
$e_n^\gamma(T)$	4.758E−1	2.453E−1	1.249E−1	6.341E−2	3.231E−2	1.667E−2
order	–	0.96	0.97	0.98	0.97	0.95
$e_p^\gamma(T)$	2.271E−1	1.194E−1	6.119E−2	3.102E−2	1.553E−2	8.021E−3
order	–	0.93	0.96	0.98	1.00	0.95
$e_R^\gamma(T)$	2.380E−1	1.228E−1	6.257E−2	3.181E−2	1.626E−2	8.439E−3
order	–	0.95	0.97	0.98	0.97	0.95
$\mathcal{C}_\gamma(T)$	1.074E−1	5.479E−2	2.764E−2	1.388E−2	6.956E−3	3.480E−3
order	–	0.97	0.99	0.99	1.00	1.00
$\mathcal{H}_{\gamma,\eta}(T)$	1.908E−1	9.237E−2	4.521E−2	2.263E−2	1.090E−2	4.964E−3
order	–	1.05	1.03	1.00	1.05	1.13

We also record the average and maximum Newton iteration counts,

$$\bar{N}_{\text{it}} = \frac{1}{K}\sum_{k=0}^{K-1} N_k, \qquad N_{\text{it}}^{\max} = \max_{0\le k<K} N_k,$$

where N_k denotes the number of Newton iterations at the k-th time step. The values in Table 2 show that the iteration counts remain moderate throughout the stiff-pressure regime.

Table 2: Newton iteration counts for the fixed-grid stiff-pressure test with $N = 6400$ and $T = 1$.

γ	10	20	40	80	160	320
$\bar{N}_{\text{it}}$	4.71	4.58	4.52	4.43	4.38	4.36
$N_{\text{it}}^{\max}$	7	6	7	8	9	9

4.3 Two-dimensional free boundary evolutions

We finally present two-dimensional simulations to illustrate free-boundary evolutions with topology changes. The simulations use the two-dimensional extension described in Appendix A. Both tests use 500×500 uniform cells on $[-5, 5]^2$, so that $h_x = h_y = 0.02$, with $\gamma = 40$ and $\tau = 10^{-3}$. The level set $n = 0.5$ is used only to visualize the numerical free boundary and is not used by the scheme. In Figures 6–7, only the central region $[-4, 4]^2$ is displayed.

Hole filling from an annular initial density. The first example starts from an annular density,

$$n_0(x, y) = 0.8\,\mathbf{1}_{\{r_{\text{in}}\le\sqrt{x^2+y^2}\le r_{\text{out}}\}}(x, y), \qquad r_{\text{in}} = 0.8, \quad r_{\text{out}} = 2.0.$$

Figure 6 shows the expansion of the outer boundary and the contraction of the inner boundary, leading to the transition from a doubly connected region to a simply connected region.

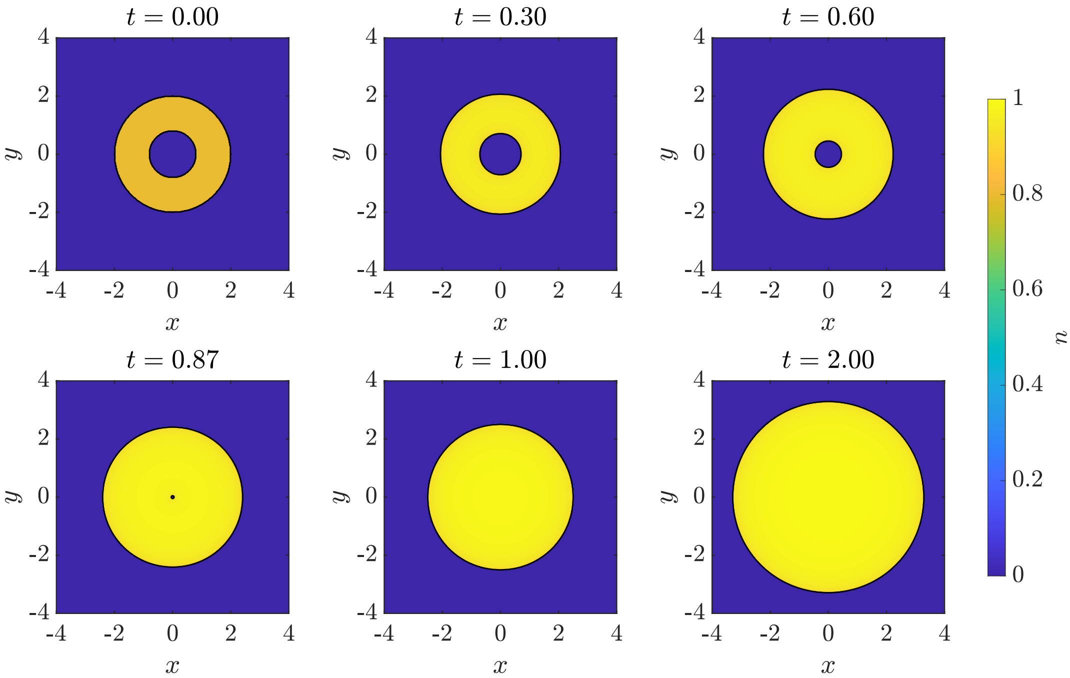

Figure 6: Density snapshots for hole filling from an annular initial density.

Merging of two tumor patches. The second example starts from two separated disks,

$$n_0(x, y) = 0.8\,\mathbf{1}_{B((-c_0,0),r_0)\cup B((c_0,0),r_0)}(x, y), \qquad r_0 = 0.8, \quad c_0 = 1.4.$$

Here $B(z, r)$ denotes the disk centered at z with radius r. Figure 7 shows that the two components first expand separately, then touch each other, and finally merge into a single connected region. After $t = 3$, the merged patch continues to expand without further topology change.

These two examples show that the fixed-grid scheme captures hole filling and component merging without explicit interface tracking.

5 Conclusion

We have developed a structure-preserving finite difference method for a pressure-driven tumor growth model with pressure-dependent proliferation. The scheme is derived from the Onsager variational principle using a modified energy shifted by the homeostatic pressure, which makes the reaction term dissipative and allows transport and growth to be treated in a unified Rayleighian formulation. The resulting fully discrete scheme uses explicit mobilities and an implicit pressure update. We proved nonnegativity preservation, the homeostatic upper bound, discrete modified energy dissipation, and a fixed-grid stiff-pressure limiting structure.

The numerical results confirm the accuracy and structure-preserving behavior of the method. The scheme captures Barenblatt profiles in the pure diffusion case, preserves mass in the no-growth case, dissipates the appropriate energy, and converges toward the Hele–Shaw limit as γ becomes large. Two-dimensional tests show that the method can handle free boundary evolutions with topology changes, including hole filling and the merging of separated patches. Future work will consider extensions to models with nutrients, chemotaxis, multiple cell populations, and adaptive resolution near free boundaries.

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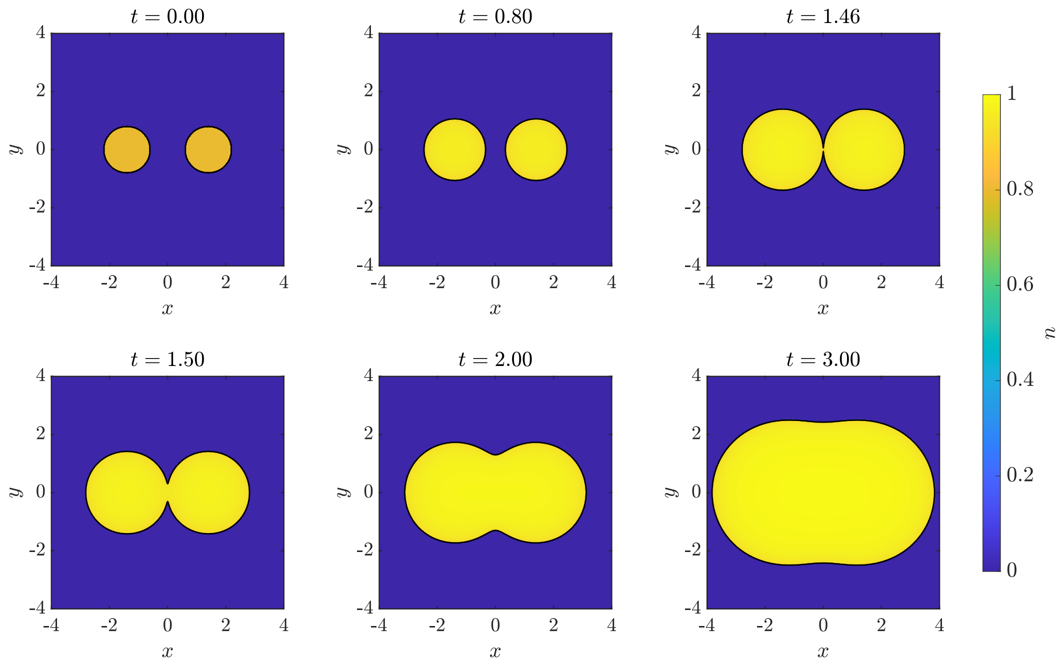

Figure 7: Density snapshots for the merging of two initially separated tumor patches.

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Data Availability The datasets supporting the current study are available from the authors on reasonable request.

Declarations

Conflicts of Interest The authors declare that they have no conflict of interest.

A Two-dimensional extension

In two space dimensions, the scheme is extended by using the same conservative finite-difference discretization in the x- and y-directions. Let h_x and h_y be the mesh sizes. We set

$$p^k_{i,j} = (n^k_{i,j})^\gamma, \qquad M^k_{i,j} = -n^k_{i,j} g(p^k_{i,j}).$$

The face mobilities are evaluated explicitly by arithmetic averages,

$$n^k_{i+\frac{1}{2},j} = \frac{n^k_{i,j} + n^k_{i+1,j}}{2}, \qquad n^k_{i,j+\frac{1}{2}} = \frac{n^k_{i,j} + n^k_{i,j+1}}{2}.$$

The pressure-gradient fluxes are

$$F^{x,k+1}_{i+\frac{1}{2},j} = n^k_{i+\frac{1}{2},j} \frac{p^{k+1}_{i+1,j} - p^{k+1}_{i,j}}{h_x}, \qquad F^{y,k+1}_{i,j+\frac{1}{2}} = n^k_{i,j+\frac{1}{2}} \frac{p^{k+1}_{i,j+1} - p^{k+1}_{i,j}}{h_y}.$$

With zero normal flux imposed on the boundary, the update is

$$\frac{n^{k+1}_{i,j} - n^k_{i,j}}{\tau} = (d_x F^{x,k+1})_{i,j} + (d_y F^{y,k+1})_{i,j} - M^k_{i,j}\left(p^{k+1}_{i,j} - p_H\right), \qquad p^{k+1}_{i,j} = (n^{k+1}_{i,j})^\gamma,$$

where

$$(d_x F^{x,k+1})_{i,j} = \frac{F^{x,k+1}_{i+\frac{1}{2},j} - F^{x,k+1}_{i-\frac{1}{2},j}}{h_x}, \qquad (d_y F^{y,k+1})_{i,j} = \frac{F^{y,k+1}_{i,j+\frac{1}{2}} - F^{y,k+1}_{i,j-\frac{1}{2}}}{h_y}.$$

The positivity preservation, homeostatic upper bound, and modified energy dissipation are obtained by the same arguments as in one dimension, using summation by parts in the two coordinate directions.

B Implementation of the nonlinear solver

This appendix summarizes the nonlinear solver used for the fully discrete scheme. At each time step, the coefficients $n^k_{i+\frac{1}{2}}$ and $\mathcal{M}^k_i$ are evaluated explicitly from the known solution, with $\mathcal{M}^k_i = -n^k_i g(p^k_i)$. The nonlinear unknown is $u = \{u_i\}_{i=1}^N$, representing n^{k+1}, and the pressure is $p_i(u) = u_i^\gamma$. The fully discrete scheme is written as

$$R_i(u) := u_i - n^k_i + \tau \mathcal{M}^k_i (u_i^\gamma - p_H) - \tau (d_h F(u))_i = 0, \qquad 1 \le i \le N,$$

where

$$F_{i+\frac{1}{2}}(u) = n^k_{i+\frac{1}{2}} \frac{u^\gamma_{i+1} - u^\gamma_i}{h}, \qquad F_{\frac{1}{2}}(u) = F_{N+\frac{1}{2}}(u) = 0.$$

We solve $R(u) = 0$ by a damped Newton method. Given $u^{(\ell)}$, the Newton correction $s^{(\ell)}$ is obtained from

$$J_R(u^{(\ell)}) s^{(\ell)} = -R(u^{(\ell)}),$$

where $J_R(u)$ is the Jacobian matrix of the residual. The Jacobian is tridiagonal in one space dimension and has the standard five-point sparse structure in two space dimensions. A backtracking line search chooses $\alpha_\ell \in (0, 1]$ so that $u^{(\ell)} + \alpha_\ell s^{(\ell)} \ge 0$ and the residual decreases sufficiently. The new iterate is

$$u^{(\ell+1)} = u^{(\ell)} + \alpha_\ell s^{(\ell)}.$$

The iteration is stopped when

$$\frac{\|R(u^{(\ell)})\|_\infty}{1 + \|n^k\|_\infty} \le \text{tol}.$$

In the computations, we use $u^{(0)} = n^k$ and $\text{tol} = 10^{-10}$. The one-dimensional Newton systems are solved by a direct tridiagonal solver, while the two-dimensional systems are solved by a sparse linear solver.

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